

of radios, alternators and voltage regulators that controlled the alternators. Engine management was the next big thing with the introduction of electronic fuel injection technology. Majority of the innovations in sophisticated engine management and transmission control have been driven by electronics through the use of MCUs, semiconductor switches and sensors. Moreover, low-cost and reliable electronics have also made body and convenience applications more efficient and standardised in modern vehicles," explains Rajeev Ramachandra, chief technology officer and co-founder, Mistral Solutions.

Application of electronics has since expanded to other vehicle systems from the days of the radio and alternator. Many sophisticated safety systems in modern cars would not have been possible without the use of electronics. In addition, the recent convergence of communication and entertainment needs have also driven rapid development of the infotainment segment in vehicles.

These developments have been made possible through the substitution of mechanical parts by electronics, more precisely, the marriage of mechanical actuator parts with electronics to develop a new area known as mechatronics, which facilitated further growth of the automotive electronics industry.

India riding the growth wave

The automotive electronics industry, a sub-set of the auto components industry in India, grew out of the offshore transfer of automotive design centres by multinational semiconductor manufacturers to leverage the country's IT expertise in design, development and simulation.

India has been seen as a base for prototyping, testing, validation and production. Global automotive original equipment manufacturers (OEMs) are also shifting automotive research and development (R&D) centres to India not only to take advantage of cost arbitrage but to also

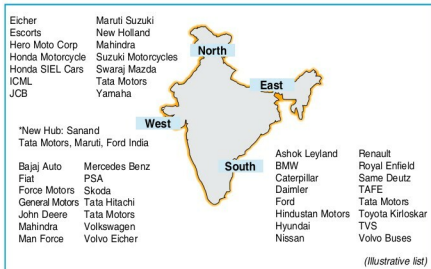


Fig. 1: Significant automotive manufacturing base of OEMs: Indian and global (Source: ACMA)

Tracing automotive electronics innovations

1970s: Introduction of electronics for engine controls

1980s: Introduction of anti-lock braking

Early 1990s: Standardisation of airbags

Late 1990s: Rapid expansion of body electronics, such as seat motors (body computers), instrument panel lighting, auto-locking systems and keyless entry

Early 2000s to date: Infotainment including sophisticated audio and video, signals sent via satellite (such as OnStar System), GPS and mapping capabilities, and satellite radio

Late 2000s: Steer-by-wire, wireless connectivity

Source: Infineon

incorporate relevant design customisations to suit the local market. Moreover, automotive electronics manufacturing is also gaining importance in the country.

The Indian auto components industry is expected to register a turnover of US\$ 66 billion by 2015-16 with the likelihood of touching US\$ 115 billion by 2020-21, depending on favourable conditions, as per the estimates of Automotive Component Manufacturers Association of India (ACMA). In addition, industry exports are projected to reach US\$ 12 billion by 2015-16 and add up to US\$ 30 billion by 2020-21.

Future growth of automotive electronics industry in India will be driven primarily by the rapidly growing passenger car market, adoption

Where does India stand in automotives

- Largest tractor manufacturer
- Second-largest two-wheeler manufacturer
- Second-largest bus manufacturer
- Fifth-largest heavy truck manufacturer
- Sixth-largest car manufacturer
- Eighth-largest commercial vehicle manufacturer

Source: ACMA

of automotive electronics, and consumer behaviour and demand.

Rapidly growing car market.

The Indian automotive industry is witnessing a phase of rapid transformation and growth, driven primarily by stable economic growth and a focus on infrastructure development. In the past two decades, the country has become a favourite destination of global OEMs for offshore transfer of production due to low production cost and increasing local demand. This, in turn, has contributed to the development of some major automo-